

SWARUP BANERJEE

Software Developer Intern

7439469690 • swarupbubai50002@gmail.com • Kolkata, India

SUMMARY

B.Tech CSE student with a solid foundation in Data Structures & Algorithms, Java, SQL, and Git. Built projects involving networking and AI concepts, with a focus on core implementation and problem-solving. Comfortable with collaborative development workflows and practical software engineering fundamentals, and eager to contribute as a software development intern.

SKILLS

Programming Languages: Java, SQL

Web Technologies: HTML, CSS, Spring Boot

Database: MySQL

Tools & Platforms: Git, GitHub

Core Concepts: Data Structures & Algorithms, OOPS, DBMS, Computer Networks, Operating Systems

PROJECTS

Network Packet Analyzer

- Built a deep packet inspection engine that parses PCAP captures at the Ethernet/IP/TCP level and extracts TLS SNI to identify applications within encrypted HTTPS traffic.
- Designed a multi-threaded processing pipeline (Load Balancer → Fast Path → Writer threads) using consistent hashing and thread-safe queues for scalable packet routing.
- Implemented flow-based traffic control with 5-tuple connection tracking, rule-based IP/app/domain blocking, and traffic breakdown reporting.

Vector Database Project

- Built a vector database engine in Java 17 + Spring Boot 3, implementing three search algorithms — Brute Force, KD-Tree, and HNSW — with a benchmarking endpoint to compare their latency.
- Designed a full RAG (Retrieval-Augmented Generation) pipeline with document chunking, semantic search, and LLM-based question answering, integrated with Ollama for embeddings and inference.
- Exposed a REST API (via Spring Boot) supporting vector insert/delete/search operations and multiple distance metrics (Euclidean, Cosine, Manhattan).

EDUCATION

B.Tech CSE, CGPA - 7.6/10

Greater Kolkata College of Engineering and Management

Expected 2027

Higher Secondary (ISC) - 91.5%

The Bhawanipur Gujarati Education Society School (ISC)

2023

CERTIFICATIONS

- Programming in Java — NPTEL
- AI in Cyber Security — EduSkills
- SQL Intermediate — HackerRank

LANGUAGES

English • Hindi • Bengali